

1.5 The Laplacian operator

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let $\Omega \subset \mathbb{R}^n$ an open domain and $f: \Omega \rightarrow \mathbb{R}$
 $x \mapsto f(x)$

a scalar field s.t. $f \in C^2(\Omega)$.

The Laplacian of f , denoted as Δf (or $\nabla^2 f$, or $D^2 f$)

is defined by

$$\Delta f(x) = \frac{\partial^2 f}{\partial x_1^2}(x) + \dots + \frac{\partial^2 f}{\partial x_n^2}(x) \in \mathbb{R}$$

1.5.1 Example

$$\begin{aligned} f: \mathbb{R}^3 \setminus (0,0,0) &\rightarrow \mathbb{R} \\ (x,y,z) &\mapsto f(x,y,z) = \frac{gmm}{r(x,y,z)} \end{aligned}$$

$$\Delta f = 0$$

1.6 Differentiation formulas

Theorem: let $\Omega \subset \mathbb{R}^n$ be an open domain,
let $f: \Omega \rightarrow \mathbb{R}$ s.t. $f \in C^2(\Omega)$, and let
 $F: \Omega \rightarrow \mathbb{R}^n$ s.t. $F \in C^2(\Omega, \mathbb{R}^n)$

then:

1) $\Delta f = \operatorname{div}(\operatorname{grad} f)$

2a) if $n=2$ we have that $\operatorname{curl}(\operatorname{grad} f) = 0 \in \mathbb{R}$

2b) if $n=3$ " " " $\operatorname{curl}(\operatorname{grad} f) = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$

3) if $n=3$ we have that $\operatorname{div}(\operatorname{curl} F) = 0 \in \mathbb{R}$

Proof:

$$1: G = \text{grad } f : \Omega \rightarrow \mathbb{R}^n, \quad G_i = \frac{\partial f}{\partial x_i}, \quad i=1, \dots, n$$

$$\text{div}(\text{grad } f) = \text{div}(G)$$

$$= \frac{\partial G_1}{\partial x_1} + \frac{\partial G_2}{\partial x_2} + \dots + \frac{\partial G_n}{\partial x_n}$$

$$= \frac{\partial^2 f}{\partial x_1^2} + \frac{\partial^2 f}{\partial x_2^2} + \dots + \frac{\partial^2 f}{\partial x_n^2}$$

□

$$2b) \quad \text{curl}(\text{grad } f) = \text{curl}(G)$$

$$= \begin{vmatrix} e_x & e_y & e_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ G_x & G_y & G_z \end{vmatrix} = \begin{pmatrix} \frac{\partial G_z}{\partial y} - \frac{\partial G_y}{\partial z} \\ \frac{\partial G_x}{\partial z} - \frac{\partial G_z}{\partial x} \\ \frac{\partial G_y}{\partial x} - \frac{\partial G_x}{\partial y} \end{pmatrix} = (*)$$

considering that $q_x = \frac{\partial f}{\partial x}$, ..., $q_z = \dots$

$$(*) = \begin{pmatrix} \frac{\partial^2 f}{\partial y \partial z} - \frac{\partial^2 f}{\partial z \partial y} \\ \vdots \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

3) curl $F: \Omega \rightarrow \mathbb{R}^3$

$$\operatorname{div}(\operatorname{curl} F) = \operatorname{div} \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}, \frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x}, \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right)$$

$$= \frac{\partial^2 F_3}{\partial y \partial x} - \frac{\partial^2 F_2}{\partial z \partial x} + \frac{\partial^2 F_1}{\partial z \partial y} - \frac{\partial^2 F_3}{\partial x \partial y} + \frac{\partial^2 F_2}{\partial x \partial z} - \frac{\partial^2 F_1}{\partial y \partial z} = 0$$

CHAPTER 2: CURVILINEAR INTEGRALS, CONSERVATIVE FIELDS, AND GREEN'S THEOREM

2.1 Curves in $\mathbb{R}^n$

2.1.1 Definitions let $n > 1$

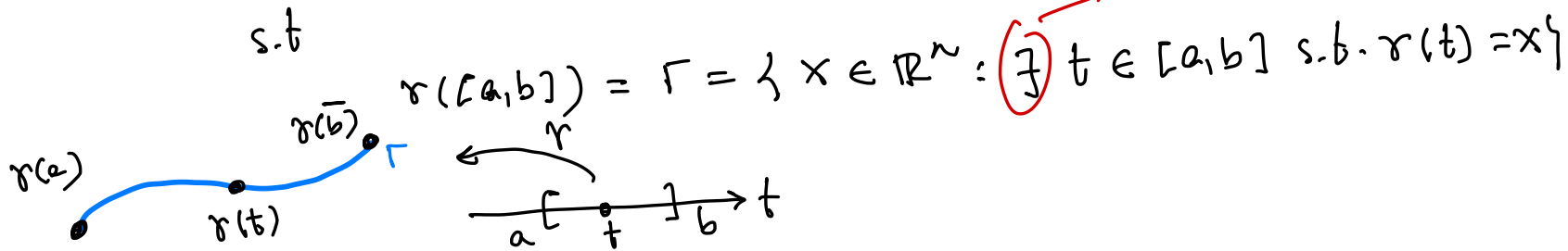
Definition 1: $\Gamma \subset \mathbb{R}^n$ is a regular simple curve if

there exists $[a, b]$ ($a < b$, an interval) and a function

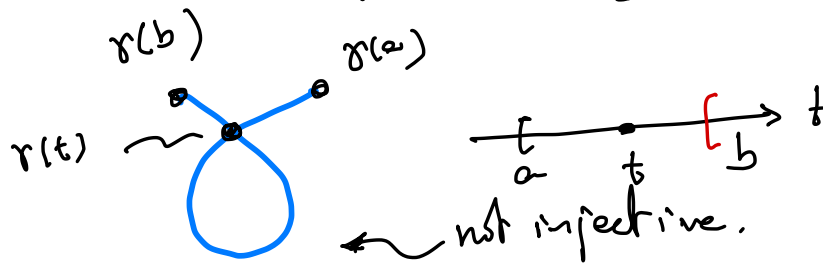
$$\gamma: [a, b] \rightarrow \mathbb{R}^n$$

$$t \mapsto \gamma(t) = (\gamma_1(t), \dots, \gamma_n(t))$$

there exist (s)

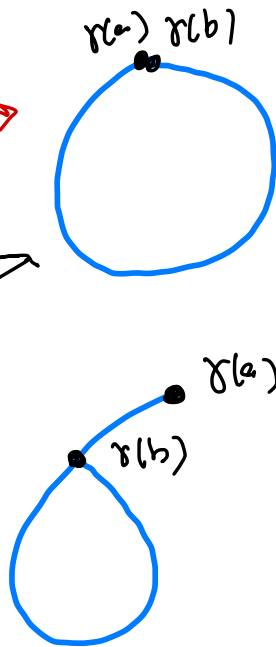


— γ is injective on $[a, b[: \forall t_1, t_2 \in [a, b[$
 with $t_1 \neq t_2 \Rightarrow \gamma(t_1) \neq \gamma(t_2)$



closed
 curve $\rightarrow$

this respects
 injectivity.



— $\gamma \in C^1([a, b], \mathbb{R}^n)$

$$- \| \gamma'(t) \| = \left(\gamma_1'(t)^2 + \gamma_2'(t)^2 + \dots + \gamma_n'(t)^2 \right)^{1/2} \neq 0$$

$\forall t \in [a, b]$
 Euclidean norm

- Definition 2: $\Gamma \subset \mathbb{R}^n$ is a closed regular simple curve if Γ is a regular simple curve and $\gamma(a) = \gamma(b)$
- Definition 3: $\Gamma \subset \mathbb{R}^n$ is a piecewise regular simple curve if $\exists \Gamma_1, \Gamma_2, \dots, \Gamma_k$ that are regular simple curves s.t. $\Gamma = \bigcup_{i=1}^k \Gamma_i$

$$\gamma_i : [a_i, b_i] \rightarrow \mathbb{R}^n$$

$$\gamma_i(b_i) = \gamma_{i+1}(a_{i+1})$$

$$i=1, \dots, k-1.$$

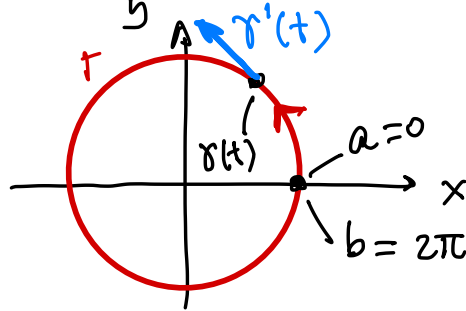
2.1.2 Examples

• Example 1:

$$\gamma : [0, 2\pi] \rightarrow \mathbb{R}^2$$

$$t \mapsto \gamma(t) = (\cos t, \sin t)$$

Γ : is a circle in $\mathbb{R}^2$, radius = 1, and centered at $(0,0)$



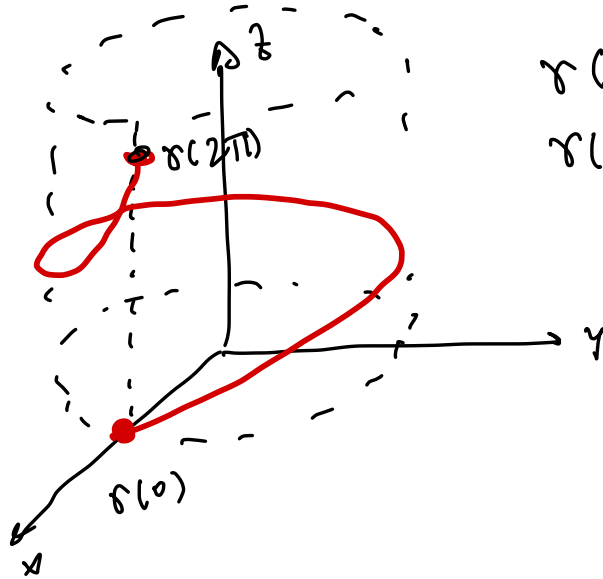
$$\gamma'(t) = (-\sin t, \cos t)$$

$$\|\gamma'(t)\| = 1 \neq 0 \quad \forall t \in [0, 2\pi]$$



• Example 2: $\gamma: [0, 2\pi] \rightarrow \mathbb{R}^3$

$$t \mapsto \gamma(t) = (3 \cos t, 3 \sin t, 4t)$$



$$\gamma(0) = (3, 0, 0)$$

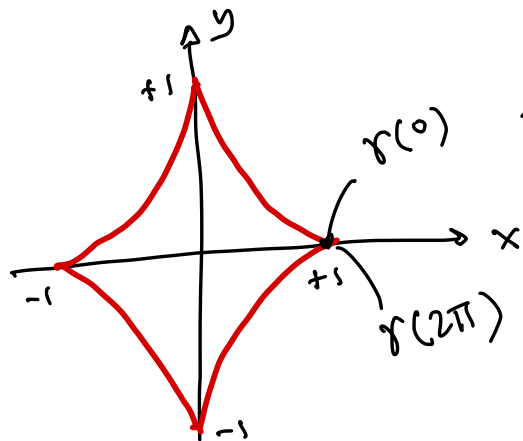
$$\gamma(2\pi) = (3, 0, 8\pi)$$

$$\gamma'(t) = (-3 \sin t, 3 \cos t, 4)$$

$$\begin{aligned} \|\gamma'(t)\| &= \left(\overbrace{9 \sin^2 t + 9 \cos^2 t}^9 + 16 \right)^{1/2} \\ &= 5 \neq 0 \quad \forall t \in [0, 2\pi] \end{aligned}$$

Example 3: $\gamma: [0, 2\pi] \rightarrow \mathbb{R}^2$

$$t \mapsto \gamma(t) = (\cos^3 t, \sin^3 t)$$



$\gamma(0)$ Γ : astroid

$$\gamma'(t) = 3(-\cos^2 t \sin t, \sin^2 t \cos t)$$

$$\in C'([0, 2\pi])$$

$$\begin{aligned} \|\gamma'(t)\| &= (9\cos^4 t \sin^2 t + 9\sin^4 t \cos^2 t)^{1/2} \\ &= 3 |\cos t \sin t| \end{aligned}$$

$$\|\gamma'(t)\| = 0 \quad \text{for } t=0, t=\pi/2, t=\pi, t=3\pi/2, t=2\pi$$

↪ non-regular simple curve

2.2 Curvilinear integrals

2.2.1 Definition

Let $\Gamma \subset \mathbb{R}^n$ be simple regular curve with parameterization:

$$\begin{aligned} \gamma: [a, b] &\rightarrow \mathbb{R}^n \\ t &\mapsto \gamma(t) \end{aligned}$$

• Def 1: Let $f: \Gamma \rightarrow \mathbb{R}$ s.t. $f \in C^0(\Gamma)$

The integral of f along Γ is defined as:

$$\int_{\Gamma} f \, dl = \int_a^b f(\gamma(t)) \|\gamma'(t)\| \, dt$$

Remark: $\text{length}(\Gamma) = \int_{\Gamma} 1 \, dl \stackrel{f=1}{=} \int_a^b 1 \|\gamma'(t)\| \, dt$
 $= \int_a^b \|\gamma'(t)\| \, dt$

• Def 2: let $F: \Gamma \rightarrow \mathbb{R}^n$
 $x \mapsto F(x) = (F_1(x), \dots, F_n(x))$

s.t. $F \in C^0(\Gamma, \mathbb{R}^n)$. The integral of F along Γ is

$$\int_{\Gamma} F \cdot dl = \int_a^b F(\gamma(t)) \cdot \gamma'(t) dt$$

• Def 3: If Γ is piecewise ^{simple} regular curve $\Gamma = \bigcup_{i=1}^k \Gamma_i$

$$\int_{\Gamma} f dl = \sum_{i=1}^k \int_{\Gamma_i} f dl \quad \text{or} \quad \int_{\Gamma} F \cdot dl = \sum_{i=1}^k \int_{\Gamma_i} F \cdot dl$$